

KSHITIJ JOSHI

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EDUCATION

The Johns Hopkins University

Baltimore, Maryland

Master of Science in Data Science

Coursework: Data Mining, Machine Translation, Algorithms for Data Science, Intro to Optimization.

Gujarat State Fertilizers and Chemicals University

Vadodara, India

Bachelor of Technology in Computer Science & Engineering with Specialization in Data Science

Coursework: Data Structures & Algorithms, Machine Learning for Intelligent Systems, Deep Learning, NLP.

TECHNICAL SKILLS

Programming Languages: Python, SQL, TypeScript/JavaScript, Rust, R

Machine Learning: PyTorch, Scikit-Learn, Hugging Face, Deep Learning, NLP, Model Training, Evaluation

LLMs & Generative AI: RAG, AI Agents, LangGraph, LangChain, RLHF, SFT, Tool Calling, MCP, LLM Evals

ML Systems & Cloud: Spark, Kafka, MLflow, Kubernetes, FastAPI, AWS, Azure, Google Cloud

Software Engineering: Git, REST APIs, Distributed Systems, Data Pipelines, Model Deployment, CI/CD

Data & Optimization: Pandas, NumPy, Statistical Analysis, Experimentation, Optimization, Analytics

WORK EXPERIENCE

EOX Vantage

Feb 2026 - Present

AI Engineer

Cleveland, OH

- Built and deployed a production **RAG platform** integrating proprietary data, semantic retrieval, and scalable LLM pipelines for enterprise workflows.
- Architected an end-to-end **multi-model LLM pipeline** for OCR, extraction, verification, and reconciliation, automating complex insurance document review.
- Partnered with stakeholders to translate business requirements into **agentic AI workflows**, integrating enterprise data, tool orchestration, and LLM evaluation.

Scale AI

Feb 2025 - Jan 2026

Machine Learning Engineer - Gen AI

Remote

- Evaluated frontier **LLMs for code reasoning**, analyzing failure modes, chain-of-thought behavior, and outputs to improve training data and eval reliability.
- Designed large-scale experiments on **RL and post-training interventions**, analyzing model behavior to inform evaluation and training decisions.

Johns Hopkins University - Center for Language & Speech Processing

Aug 2024 - Jan 2025

Research Assistant

Baltimore, MD

- Researched **low-resource neural machine translation** using LLM-assisted methods, improving Gujarati-English Flores200 performance by **15%**.
- Optimized **MarianMT** and **NLLB** through training and error analysis, reducing translation errors by **20%**.

Johns Hopkins University - Advanced Robotics (ARCADE) Lab

Jan 2024 - Jan 2025

Research Assistant

Baltimore, MD

- Developed **ML models** for real-time cognitive load estimation in telerobotic surgery using pupillometry and brightness-aware feature engineering.
- Applied gaze entropy and **Random Forest** to physiological signals, reducing misclassification by **20%**.

MarwizTech

Jan 2023 - Aug 2023

Machine Learning Engineer

Vadodara, India

- Built and deployed **CNN-based vision models** with Docker, FastAPI, and REST APIs, achieving **98% accuracy** while reducing inference latency by **25%**.

KEY PROJECTS

Financial Document Intelligence with Retrieval-Augmented Generation

Fall 2024

- Built an end-to-end **RAG and evaluation pipeline** using **Llama 3.2**, **LangChain**, and **FAISS**, combining query rewriting, hybrid retrieval, and **MMR** for document reasoning.
- Engineered a custom **HTMLTableAwareTextSplitter** preserving structured tabular context, improving retrieval accuracy by **35%** on financial reports.

LegalEase: Neural Machine Translation for Legal Documents

Spring 2024

- Fine-tuned and evaluated **IndicTrans** with **Fairseq** for domain-specific legal translation, achieving a BLEU score of **0.58**.
- Built a scalable **NLP inference pipeline**, reducing manual translation effort by **40%** through automated document processing.

PUBLICATIONS

- K. Joshi, S. Rana, et al., "Evaluation of Artificial Intelligence Methods for Fracture Detection in Orthopedic X-Rays", **Rapid Fire Presentation Award**, Clinical Orthopedic Society (Under Review)
- K. Joshi, A. Vyas, et al., "Cognitive-Chair: AI-Based Advanced Brain Sensing Wheelchair for Paraplegic/Quadriplegic People", **Best Paper Award**, AIST 2022; IEEE Xplore, 2023 [Paper]